Biochar for Urban Tree Plantings

[Bracketed text in red should be adjusted or removed per region and project goals. Guidance for biochar-specific decisions is provided in the Biochar in Stormwater BMPs Guidebook for Stormwater Practitioners (LINK).]

PART 1 – GENERAL

Purpose

For incorporating biochar into new urban tree planting to support plant growth in traditionally harsh environments.

Description

Incorporating biochar into new urban tree plantings including the furnishing, soil preparation and amending of biochar. This specification covers the use of biochar for supplementing in-situ soils for new tree plantings.

Related Specifications

Biochar Material and Quality Control
Biochar Material Handling
Biochar Material Stockpiling
Biochar-Amended Compost

References

Adhikari et al., 2022. Optimizing water holding capacity and hydrophobicity of biochar for soil amendment – A review (LINK)

Chowdhury, 2021. Biochar Amendment for Enhanced Infiltration and Aggregation of Compact Urban Soils (LINK)

Edeh et al., 2020. A meta-analysis on biochar's effects on soil water properties – New insights and future research challenges (LINK)

Fields-Johnson, Chris, and Chelsi Abbot. Applications of Biochar for Arboriculture. International Society of Arboriculture (ISA).2020. (LINK),

Urban Tree Foundation. Planting details and specifications (LINK)
Washington D.C. Department of Energy and Environment (DOEE). *3.14 Tree Planting and Preservation*. (LINK)

Submittals

1. Submit per requirement under [General Provisions, Section XXX, ...]

2. Testing results from in-situ soil test, including particle size distribution, organic matter content, bulk density, saturated hydraulic conductivity, and infiltration rate.

PART 2 – MATERIALS

Biochar

Refer to the following specifications

1. Biochar Material and Quality Control

Biochar will be applied to the soil at a rate of [5% to 20% v/v, …].

Biochar-Amended Compost

Refer to the following specifications

1. Biochar-Amended Compost

Biochar can be applied as a compost amendment if compost is determined needed as an additional soil amendment.

PART 3 – EXECUTION

Biochar Application

1. Biochar soil amendments may be added to new tree plantings in urban settings to improve soil retention and nutrient retention qualities. Biochar soil amendments are just one of several soil amendments that may be specified for tree planting soil improvements. Follow all soil amendment specifications applicable to each planting application.
    a. Biochar soil amendments are most compatible with dry, sandy, well-drained soils, and/or soils with low nutrient content. To determine the existing soil type of proposed tree planting sites, use one of the following methods:
    b. Take a soil sample and send it to a lab or cooperative extension for testing. Resulting soil texture analysis, nutrient analysis, and organic matter analysis will help determine if a biochar soil amendment would be beneficial. Soils with limited available nutrients and low organic matter can be amended with a combination of biochar and organic matter (if specified by design). Loamy-sand, sand-loam, and sandy clay loam can be amended with [5-20%] biochar by volume.
    c. Use the USDA NRCS Web Soil Survey to determine Hydrologic Soil Group for the tree planting site. Sites with Class A, B and C soils are suitable for biochar soil amendments. Local soil experts should be consulted for amending HSG D soils with biochar for best results.
        i. Class A Soils: 5-20% biochar by volume
        ii. Class B or C Soils: 5-10% biochar by volume

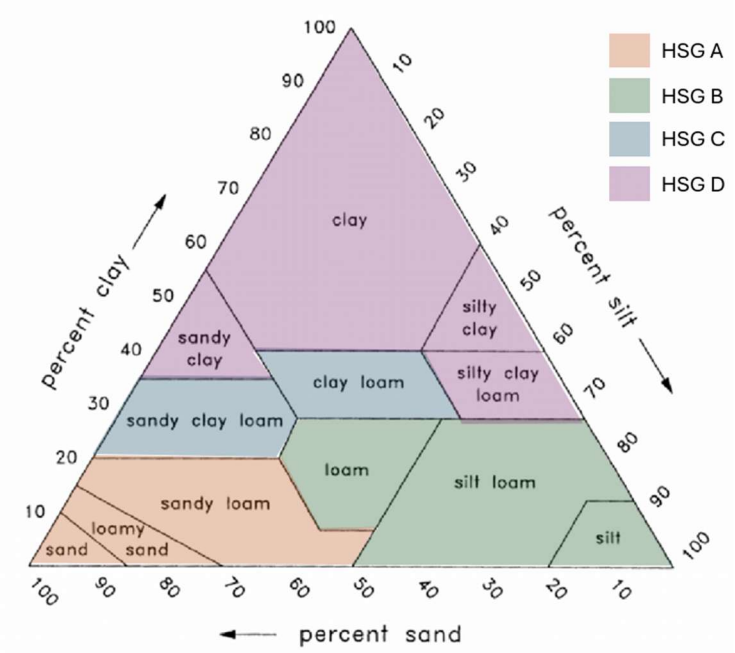


Figure 1. Soil texture and hydraulic soil group based on particles size distribution

2. A field analysis may also be used to determine if the existing soil type is dry, coarse textured, well- to excessively well-draining, or disturbed. Biochar soil amendments are suitable for these soil types because it will increase moisture and nutrient retention.
    a. Add no more than 20% biochar by volume
3. Biochar amendments shall be mixed uniformly into existing soil prior to planting. Excavate tree planting holes and retain removed soil to mix biochar amendments in. See "Mixing Amendments and Preparing Backfill", below.

Mixing Amendments and Preparing Backfill

1. For new tree plantings, a minimum of 50% of the backfill soil shall come from the site itself to avoid shock to the tree during the establishment period (Fields-Johnson and Abbot, 2020). Protect existing soil from compaction, contamination, and degradation during the installation process.
    a. It is recommended to add existing site soil to pre-mixed biochar or biochar/compost blend to reduce shock to tree during establishment.
2. Biochar and existing soil (and any other specified soil amendments, i.e. compost) can be mixed together on-site using a variety of tools and methods including, but not limited to:
    a. Hoppers and Conveyors for soil blending
    b. Batch Mixers
    c. Concrete Mixer
    d. Paddle Mixer and bucket
    e. Wheelbarrow and shovel

Installation

Follow local jurisdiction's tree planting specifications for planting site preparation and tree planting. The following specifications refer only to those soils being amended with biochar during new tree planting activities.

1. Locate and dig tree-pit as specified on plans.
2. Save soil removed from the tree pit as backfill for the tree planting. This backfill will later be mixed with biochar before applied to the tree hole. See "Application" section, above.
3. Place the tree in the pit so that the root collar is exposed just above finished grade, about 2-3 inches.
4. Backfill the tree with amended soil, taking care not to cover the root collar.
5. Stake or stabilize the tree as necessary. Follow municipal or jurisdictional guidelines.
6. Provide adequate watering to trees as necessary during their establishment period.
7. Follow local regulations for mulch application after new tree planting.